

UNESCO 오픈사이언스 권고 이행 국가보고서 취합본

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1. 오픈사이언스에 관한 공통의 이해 증진

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1 2021년 권고가 해당 부처와 기관, 유관 기관 등에 의해 장려되고 공유되어 왔는가?

Has the 2021 Recommendation been **promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations** in your country?



Yes. The relevant ministries, the Ministry of Foreign Affairs and the Ministry of Science and ICT (MSIT) have been directly involved in the preparation of the Recommendation, together with the Korean National Commission for UNESCO. After the Recommendation was endorsed, the Ministry of Science and ICT (MSIT), the Korean National Commission for UNESCO, the Science and Technology Policy Institute (STEPI), the Korea Institute for Science and Technology Information (KISTI), the National Library of Korea, the National Assembly Library, Korea Education and Research Information Service (KERIS), and other relevant organizations, experts and stakeholders in Korea have been promoting the dissemination and implementation of the Recommendation.

1.2 2021년 오픈사이언스 권고가 귀하의 국가 언어로 제공되는가?

Is the 2021 Recommendation on Open Science **available in the national language(s)** of your country? **YES/NO** If yes, which language(s)?



Partly yes. In late 2020, the Korean National Commission for UNESCO published and distributed a Korean booklet, "Towards a UNESCO Recommendation on Open Science," which introduces the final draft UNESCO Recommendation and the process of developing the draft to the public.

REFERENCE:

A Link to the Korean booklet, "Towards a UNESCO Recommendation on Open Science (유네스코 오픈사이언스 권고를 향하여)" : <https://unesco.or.kr/wp-content/uploads/2024/06/유네스코-오픈사이언스-권고를-향하여.pdf>

1.3 귀하의 국가에서 자국 단체나 기관에 의해 오픈사이언스에 관한 **인식 제고 활동**이 이루어졌거나 혹은 2025년 말까지 이루어질 예정인가?

Have there been **awareness raising activities on open science**, including all its key elements, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.q, 16.h)



Yes. In 2021 and 2022, the Korean National Commission for UNESCO produced and distributed newsletters and videos about the Open Science Recommendation online to inform the public about the Recommendation. Since 2022, the Science and Technology Policy Institute (STEPI) has regularly organized a domestic stakeholder meeting for the implementation of the Recommendation, with the support of the Ministry of Science and ICT (MSIT). Moreover, the National Library of Korea holds an annual open access conference, Open Access Korea (open access) repository workshops for policymakers, researchers, libraries, and academic societies. The Korea Institute for Science and Technology Information (KISTI) also raises awareness of open science through the annual Future Research Information Forum and periodical AccessOn newsletters. KISTI published a guidebook on open access publishing in 2021 and 2022 and organized the Transformative Agreement Working Group from 2022 to 2023.

The "National Open Access Policy Forum" took place in June 2021, which was followed by the "Joint Declaration on Open Access" in August 2021. The Declaration was signed by major research funding agencies and public scholarly information service organizations, including the National Library of Korea, the National Assembly Library, the National Research Foundation of Korea (NRF), the Korean Federation of Science and Technology Societies (KOFST), the Korea Institute of Science and

Technology Information (KISTI), and the Korea Education and Research Information Service (KERIS). Later, initiated by the National Library of Korea and KISTI, 12 organizations related to open access (public institutions, research institutes, universities, and academic institutions) made an "OA Declaration of Action" in 2023 and opened the Declaration webpage that enables any interested parties to join the Declaration afterward. As of 2024, a total of 23 organizations have signed the Declaration of Action.

REFERENCE:

National Library of Korea OA Declaration of Action (Korean),
<https://www.oak.go.kr/about/aboutOak.do?menuSeq=97>

1.4 공적으로 지원한 연구에 오픈사이언스의 가치와 원칙을 반영하기 위하여 귀하의 국가에서 이미 시행했거나 2025년 말까지 시행을 계획 중인 일이 있는가?

Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of **open science, in publicly funded research?** (ref.: (i) 16.a. 16.b. 16.c. 16.d. 16.e. 16.h)



Yes. Multiple national policies, plans, and initiatives on publicly funded research incorporate the principles and values set out in the Open Science Recommendations – openness, collaboration, collective benefits, inclusion, transparency and responsibility. The 4th Basic Plan on the Management and Utilization of Research Outputs (2021-2025) is one such example. Under the plan, the Korean government promotes open science by encouraging publicly funded research results to be widely disseminated and shared. Specifically for research data, the "Strategy for Sharing and Utilizing Research Data to Promote Innovative Growth" was devised in 2018, and a data management plan started to be developed in a national R&D project. In addition, national-level research data platforms, such as the Korea Bio-DataStation (K-BDS) and the Korea Materials Data Station (K-MDS), were established and are currently operating to support field-specific data sharing.

2. 오픈사이언스 정책의 개발

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 귀하의 국가는 과학기술혁신에 관한 국가정책이나 전략, 실행계획을 가지고 있는가?

Does your country have a **national policy, strategy or plan of action on science, technology and innovation (STI)**?



Yes. There is a five-year basic plan for science and technology and a national medium- and long-term investment strategy for research and development. Currently, the 5th Basic Plan on Science and Technology (2023-2027) and the first National Strategy on Mid- to Long-Term R&D Investment (2023-2027) are being implemented while being reviewed and updated annually. Moreover, the current presidential agenda recognizes the importance of science, technology and innovation (STI).

위의 계획이 다음의 오픈사이언스 관련 항목을 포함하는가?

- does it include any of the following open science elements:
 - o open access to scientific knowledge, o open science infrastructure
 - o open engagement of societal actors, o open dialogue with other knowledge systems



Yes. The recently established 5th Basic Plan on Science and Technology (2023-2027) emphasizes promoting access to and utilization of scientific and technological achievements. It also highlights the contribution of science and technology to solving social problems and national challenges. As specific tasks, the Basic Plan proposes "Building an open and innovative research environment (1-2-3)" and "Revitalizing science culture with public engagement (1-5)". In addition, it calls for specific actions to develop a digital research environment, including open science infrastructure, to improve research performance and accessibility, to expand openness in R&D, and to encourage public participation in science.

REFERENCES :

The Ministry of Science and ICT (December 14, 2022) The 5th Basic Plan on Science and Technology (2023-2027), The Deliberative Council of the Presidential Advisory Council on Science

& Technology (PACST), Link to the Korean webpage,
<https://online.fliphtml5.com/lukuo/mlhr/#p=1>

2.2 귀하의 국가에서 2021년 오픈사이언스 권고에서 제시한 원칙과 가치, 정의와 연계된 사항이나 오픈사이언스를 다룬 국가 차원의 정책과(혹은) 법제가 있는가?

In your country, is there **a policy, or a set of policies and/or legal framework(at the national level that address open science** or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

YES / PARTLY /NO



Partly yes. The Framework Act on Science and Technology, which is the backbone law that guides science and technology policy in the Republic of Korea, affirms the key values and principles of the UNESCO Open Science Recommendation, such as openness, collaboration, collective benefits and responsibility. Specifically, Article 5 of the Framework Act requires the government to actively support "creative research and development (R&D) and open innovation activities in the fields of science and technology." In general, the Framework Act emphasizes science-society linkages and cooperation in science, technology and innovation (STI).

On top of that, legislative efforts have been underway since 2023 to promote the management and utilization of research data at the national level. With the opening of the 22nd National Assembly in 2024, three bills have been proposed for the "Act on Promotion of National Research Data Management and Utilization."

REFERENCES :

The Framework Act on Science and Technology (first enacted in 2001 and last updated in 2024) :

A Link to the Korean version of the Framework Act (official webpage in Korean run by the Korean Law Information Center),

<https://law.go.kr/%eb%b2%95%eb%a0%b9/%ea%b3%bc%ed%95%99%ea%b8%b0%ec%88%a0%ea%b8%b0%eb%b3%b8%eb%b2%95>

A Link to the translated version of the Framework Act (unofficial site in English, operated by the

Korean Legislation Research Institute),

<https://law.go.kr/lslInfoP.do?lsiSeq=165472&lsId=009177&chrClsCd=010202&urlMode=engLslInfoR&viewCls=engLslInfoR#0000>

2.3 귀하의 국가에는 오픈사이언스를 진흥시키는 별도의 정책수단이 존재하는가?

In your country, are there **specific policy instruments that aim to promote open science**?

YES/ UNDER DEVELOPMENT/NO



Yes. With relevant policy measures and projects, the Korean government is promoting the sharing and utilization of research data at the national level. The Ministry of Science and ICT (MSIT) devised the Strategy for Sharing and Utilizing Research Data in 2018. In line with the Strategy, the National Research Foundation of Korea (NRF) began to require principal investigators of selective national R&D projects to develop data management plans (DMPs) in addition to project proposals. As research data infrastructure, data stations were established in the fields of bio-research and materials research, and a national research data platform service was also launched by the Korea Institute for Science and Technology Information (KISTI). The establishment and sustainable operation of a national research data platform was also declared a presidential agenda. As of 2024, the legislative bill, the "Act on Promotion of National Research Data Management and Utilization," is under review in the National Assembly.

REFERENCE:

The Ministry of Science and ICT (January 19, 2018) Strategy for Sharing and Utilizing Research Data to Promote Innovation Growth, The Steering Committee of the National Science and Technology Deliberative Council.

2.4 귀하의 국가에서 2021년 권고에 맞추어 국가 차원의 오픈사이언스 이행을 위하여 마련된 계획이나 전략, 정책, 로드맵이 있는가?

In your country, is there a **national implementation plan, strategy, policy or roadmap for implementation of open science at the national level** in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

YES/ UNDER DEVELOPMENT /NO



Partly yes. There is no single national strategy for open science, but there are national plans and strategies that promote specific open science practices and build up open science-friendly environments. As mentioned above in Question 2.1, the 5th Basic Plan on Science and Technology (2023-2027) promotes the creation of digital research infrastructure, access to public research results, and the advancement of science communication and culture. In addition, the 4th Basic Plan on the Management and Utilization of Research Outputs (2021-2025) sets a goal of an open research and innovation ecosystem in which research outputs are easily accessed, well-managed and fully utilized; it specifies required actions to enhance open research ecosystem.

REFERENCE:

The Ministry of Science and ICT (February 25, 2021) The 4th Basic Plan on the Management and Utilization of R&D Outputs (2021-2025), The Deliberative Council of the Presidential Advisory Council on Science & Technology (PACST), Link to the webpage (in Korean), <https://online.fliphtml5.com/lukuo/mqzx/#p=1>

2.5 귀하의 국가에서 2021년 권고에 맞추어 연구수행기관이나 대학, 학회나 학술원 등과 같은 기관 차원에서 오픈사이언스를 촉진하기 위하여 마련된 정책이나 전략이 있는가?

In your country, are there **policies and/or strategies that promote open science** in line with the 2021 Recommendation **at institutional level**, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d. 17.e. 17.g)

YES /UNDER DEVELOPMENT/ NO



Yes. Since 2014, the National Library of Korea has supported research organizations, schools, and public institutions to establish open access repositories. As a result, a total of 61 institutional open access repositories were developed and are in operation as of 2023. The National Library of Korea also established an open access policy in line with the UNESCO Recommendation and has been supporting other organizations in developing their own open access policies. Eight Korean institutions (National Library of Korea, Daegu Gyeongbuk Institute of Science and Technology, Ajou

University Central Library, Yonsei University Medical Library, Jeonbuk State Institute, Korea Advanced Institute of Science and Technology (KAIST), Korea Institute for Health and Social Affairs (KIHSA), and Korea Institute of Ocean Science & Technology) have developed their own open access policies which are being enlisted in the international Registry of Open Access Repository Mandates and Policies (ROARMAP) (<https://roarmap.entities.org/1948/>). Since 2018, the Korea Institute of Science and Technology Information (KISTI) has also developed and implemented its open access policy registered in the international ROARMAP.

Meanwhile, the National Research Foundation of Korea (NRF) and the Korea Federation of Science & Technology Societies (KOFST) have provided financial support to domestic academic societies for journal publishing through the project (i.e., Domestic Academic Organization Support Project) funding from the Ministry of Education. The project evaluation indicators include "whether the journal is free to access (or open access)," which eventually facilitates domestic society-owned journals to adopt a Diamond OA model. Currently, 1,257 out of 2,831 journals enlisted in the KCI Academic Database of the National Research Foundation of Korea (NRF) provide full access to their original content without charge. In addition, it is becoming more common for a research organization, such as the National Institute of Ecology, to publish its organizational journals as open access.

Selective public research organizations are also equipped with data sharing platforms. The National Institute of Ecology operates the EcoBank platform that gathers and distributes ecological data, and the Korea Institute of Geoscience and Mineral Resources (KIGAM) runs the Geo Open Big Data Platform. Plus, KIGAM and some other public research organizations have their own research data policies to manage research data at the institutional level.

REFERENCES:

Korea Advanced Institute of Science and Technology (KAIST)'s Open Access Policy (in English): <https://koasas.kaist.ac.kr/about/policy>

Korea Institute for Health and Social Affairs (KIHSA)'s Open Access Policy (in English): <https://repository.kihasa.re.kr/en/about?t=4>

The National Institute of Ecology, EcoBank platform (in Korean): <https://www.nie-ecobank.kr/cmmn/Index.do>

The Korea Institute of Geoscience and Mineral Resources (KIGAM), Geo Open Big Data Platform (in Korean): <https://data.kigam.re.kr/>

2.6 귀하의 국가에서 국가 차원의 오픈사이언스를 촉진하기 위하여 마련된 별도의 펀딩메커니즘이 있는가?

In your country, are there **specific funding mechanisms for open science** at the national level? (ref.: (ii) 17.i. (iii) 18.i. (iv) 19. c)

YES / UNDER DEVELOPMENT/ NO



No. There is no dedicated single funding mechanism exclusively for open science. However, there are a variety of government funding programs in which the Korean government supports specific open science practices and R&D activities that reflect the core principles of the UNESCO Open Science Recommendation.

2.7 귀하의 국가에서 오픈사이언스 활동이 기존 연구펀딩 메커니즘에 고려·통합되어 있는가?

In your country, have **open science practices been integrated in existing research funding mechanisms**? YES/NO



Yes. The core values of the UNESCO Open Science Recommendation and specific open science practices are supported by the existing research funding mechanisms. The Korean government supports national R&D programs and public research institutions relevant to open science to facilitate the sharing and utilization of research data and public engagement in science.

In addition, when the results of a national research and development (R&D) project are published as open access articles, the article processing charge (APC) can be reimbursed from the project budget.

As the Republic of Korea joins the Pillar 2 program of EU Horizon Europe as an associate member in 2024, Korean participants of the EU program are expected to comply with the Horizon Europe requirements on open access publishing and research data management.

2.8 귀하의 국가에서 2021년 권고에 맞추어 형평성있는 **오픈사이언스 민관협력**을 촉진하기 위하여 노력해왔는가 혹은 2025년 말까지 노력할 예정인가?

Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster **equitable public-private partnerships on open science** in line with the 2021 Recommendation? (ref.: (ii) 17.i) YES/ NO



Yes. Public-private partnerships are being fostered in the process of implementing open science. Not only government entities but also academic societies, publishers, private organizations, and societal actors have participated in the implementation of the UNESCO Open Science Recommendation.

For alternative transformative agreements with global publishers, the Korea Institute of Science and Technology Information (KISTI) organized the Transformative Agreement Working Group, in which relevant domestic stakeholders got together and explored available options. It began after KISTI signed a Memorandums of Understanding (MoUs) with the Korea University & College Library Association (KUCLA) and the Korea Special Library Association (KSLA) in 2021. In fact, 25 government-funded research institutes under the National Research Council of Science & Technology (NST) made transformative agreements with two major global publishers (Elsevier and Wiley) in 2021. Since then, the number of transformative agreements made with global publishers (Cambridge University Press, Royal Society of Chemistry, Karger, Company of Biologists, Institute of Physics, ACM, etc.) has increased in Korea.

For domestic journals, the National Library of Korea, in cooperation with the National Research Foundation of Korea (NRF), established the Korea Journal Copyright Information (KJCI) system in 2017; it helps share information on copyright policies of domestic journals, provides guidance and counseling on open access publishing to academic societies, and supports the conversion of journal content into a XML format. As a result, as of 2024, more than 15,000 articles from 128 domestic journals are available on the web in a JATS XML format.

Plus, for science education and communication, the KAOS Foundation has developed video content in the field of basic science and delivered it to the public through KISTI's ScieonceOn platform, based on a business agreement signed with KISTI in 2024. A citizen science organization, Nature Ring, has strengthened its collaborations with researchers from national research institutes, such as the National Institute of Ecology and the National Institute of Forest Science.

REFERENCES:

The Korea Journal Copyright Information (KJCI) system (in Korean):
<https://copyright.oak.go.kr/main.do>

The Korea Institute of Science and Technology Information (KISTI)'s ScienceOn (in Korean):
<https://scienceon.kisti.re.kr/main/mainForm.do>

The KAOS Foundation (in Korean): <https://ikaos.org/kaos>

Nature Ring (in Korean): <https://www.naturing.net/landing>

2.9 귀하의 국가에서 **오픈사이언스에 관한 국가 모니터링 프레임워크**가 존재하는가? 혹은 국가의 STI 정책 평가·점검 프레임워크에 오픈사이언스 관련 특정 지표가 포함되어 있는가?

Is there a **national monitoring framework for open science** in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy? (ref.: (ii) 17.i. 23.c)

YES/ UNDER DEVELOPMENT/ NO



Partly yes. Although there is no comprehensive framework for monitoring open science activities at the national level, there exist monitoring frameworks targeted at specific open science activities, such as open access publishing. The Korea Institute of Science and Technology Information (KISTI)'s Open Access Monitor Korea informs us of global open access publishing trends. The Korea Journal Copyright Information (KJCI), run by the National Library of Korea, provides information on copyright policies of domestic journals, as well as their open access (OA) publishing status (e.g., Gold OA and Green OA). At the same time, KISTI's DataOn Statistics summarizes the status and characteristics of research data uploaded in DataOn.

REFERENCES:

The Korea Institute of Science and Technology Information (KISTI)'s Open Access Monitor Korea (in Korean): <https://www.ksli.or.kr/oamk/main/main.do>

KISTI's DataOn Statistics (in Korean), <https://dataon.kisti.re.kr/datause/datauseStatus.do>

The Korea Journal Copyright Information (KJCI) system (in Korean):
<https://copyright.oak.go.kr/main.do>

3. 오픈사이언스 인프라 및 서비스 투자

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1 귀하의 국가에서 **국내총생산(GDP) 중 연구개발지출(투자)이 차지하는 비중**은 얼마인가?(가장 최근 공식 데이터 기준) 해당 데이터의 출판연도와 생산연도를 표시하라.

What is the latest official data for your country's **percentage of national gross domestic product (GDP) dedicated to research and development expenditure**? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)



According to Statistics Korea, the share of R&D expenditure to Gross Domestic Product (GDP) is 5.21% as of 2022, which is slightly higher than 4.93% of GDP reported in 2021 at the UNESCO UIS.

REFERENCES:

Statistics (last updated on April 25, 2024) Share of R&D expenditure to GDP, Link to the webpage (in Korean), <https://www.index.go.kr/unify/idx-info.do?idxCd=4206>

UNESCO UIS – Science, Technology and Innovation: GERD as a percentage of GDP (Republic of Korea), <https://data.uis.unesco.org/>

3.2 귀하의 국가에서 **전체 인구 중 신뢰할만한 인터넷·광대역에 접근할 수 있는 비중**은 몇 퍼센트인가? (가장 최근 공식 데이터 기준) 해당 데이터의 출판 연도와 생산 연도를 표시하라.

What is the latest official data for **the percentage of your country's population that have access to reliable internet and bandwidth**? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)



As of 2023, 94% of the national population (population aged 3 and over) are Internet users, and 82.5% of households have access to Internet, according to Statistics Korea. According to ITU data, 97.4% of the national population (population aged 14 and over) have internet access, and 46.6%

subscribed to fixed broadband services as of 2023.

REFERENCES:

Statistics Korea (last updated on May 1, 2024), Share of Households with Internet Access, Link the webpage (in Korean): https://www.index.go.kr/unity/potal/main/EachDtlPageDetail.do?idx_cd=1345

Statistics Korea (last updated on May 1, 2024), Share of Internet Users, Link the webpage (in Korean): https://www.index.go.kr/unity/potal/main/EachDtlPageDetail.do?idx_cd=1346

ITU, DataHub – Korea (Rep. of): Connectivity, <https://datahub.itu.int/>

3.3 귀하의 국가에 **활성화된 국가연구·교육네트워크(NRENs)**가 존재하는가?

Are there **national research and education networks (NRENs) active** in the country? (ref.: {jjjl 18.c)

YES/ NO



Yes. Edunet, a national education network, provides one-stop access to open education resources (OER) produced by 17 Offices of Education for elementary and middle school teaching and learning. Since 1996, Edunet has been operated by the Korea Education and Research Information Service (KERIS) with financial support from the Ministry of Education.

KERIS also offers the Korea Open CourseWare (KOCW) service, which provides online courses created by Korean universities and continuing education institutions. In addition, the K-MOOC (Korea Massive Open Online Course) service is run by the National Institute for Lifelong Education. The National Library of Korea also provides open access training sessions to librarians and repository managers working in research organizations, schools, and public institutions through its education platform.

Since 1998, KERIS has been operating the Research Information Sharing System (RISS), in which domestic researchers can search for and get access to the information on theses and dissertations of domestic universities, academic literature held by domestic libraries, and scholarly information acquired from overseas.

Moreover, several networks for research exist. For instance, the Korea Institute of Science and Technology Information (KISTI) operates the Global Science Experimental Data Hub Center (GSDC)

that supports access to the large amounts of experimental data produced by the world's largest particle accelerators (CERN, KEK) and giant observatories (LIGO). KISTI also provides the Korea Research Environment Open NETwork (KREONET) service. KREONET, a network for domestic researchers in the field of science and technology, is capable of transmitting a minimum of 200 Gbps and a maximum of 1.2 Tbps per section; it is based on 18 regional network centers nationwide and connected to the international research network through GLORIAD; it provides stable Internet services to about 200 network subscribers through IX at home and abroad.

REFERENCES:

The Korea Education and Research Information Service (KERIS)'s Edunet (in Korean): <http://www.edunet.net>

The National Library of Korea, Education websites for Librarians (in Korean), <https://edu.nl.go.kr>

KERIS' the Korea Open CourseWare (KOCW) (in Korean): <http://www.kocw.net>

The National Institute for Lifelong Education, K-MOOC (in Korean): <http://www.kmooc.kr>

KERIS' the Research Information Sharing System (RISS) (in Korean): <http://www.riss.kr>

KISTI's KREONET (in Korean): <https://kreonet.net>

3.4 귀하의 국가에 국가 혹은 기관 차원의 오픈사이언스 인프라가 존재하는가?

Are there **national or institutional open science infrastructures** in your country? (ref.: (iii) 18. d, 18.e. 18.k. 18.l, 18.m) **YES/ UNDER DEVELOPMENT/ NO**



Yes. There are many forms of digital infrastructure that support open science. First, the National Science and Technology Information Service (NTIS), designed to provide information on science and technology and national research and development projects, has operated since 2008. NTIS is regarded as a well-performing core facility that manages and disseminates R&D information at the national level, as stipulated in the Framework Act on Science and Technology; it has been awarded by diverse entities, including the UN Public Service Award in 2012.

In addition, distinctive digital infrastructures serve specific open science activities, such as open access publishing and research data sharing. The Korea Institute of Science and Technology

Information (KISTI) operates the AccessOn platform to support open access publishing. Through the platform, domestic research institutions and academic societies can not only search and utilize open access articles but also publish them. The Korea OA Journal service provided by the National Research Foundation of Korea (NRF) enables users to search for and utilize all the open access contents from KCI-registered journals at once. The National Library of Korea, in charge of the Open Access Korea (OAK) project, has developed the OAK Repository Portal since 2014, which enables an integrated search for OAK (open access) repositories.

To share and utilize research data, the Korea Institute of Science and Technology Information (KISTI) established the national research data platform, DataOn in 2020. DataOn allows us to search and utilize available research data in a unified manner. KISTI also developed research data management software, NaRDA, and has distributed it to domestic institutions; as a result, 23 public institutions have established their own repository systems based on NaRDA. In the field of bio-research, the Korea Bio Data Station (K-BDS) has been operating since 2022, and the Korea Materials Data Station (K-MDS) is in operation in the field of materials science.

For open collaboration, KISTI has set up a platform, ScienceOn. The contents and services of ScienceOn are delivered via an effective API Gateway, such as OpenAPI and stOAI. Until early 2024, a total of 253 organizations (domestic research institutes, universities, public institutions, and companies) have used the ScienceOn API Gateway.

In addition, the Korea DOI (Digital Object Identifier) Center, run by KISTI, provides an essential digital infrastructure service; it assigns a permanent unique identifier to digital content so that the content itself can be identified, searched, found and used with the identifier. As of August 2024, about 26 million DOIs have been assigned to publications and scientific data that are produced by about 1,600 organizations in Korea.

REFERENCES:

The National Science and Technology Information Service (NTIS) (in Korean), <https://www.ntis.go.kr>

AccessOn, operated by the Korea Institute of Science and Technology Information (KISTI) (in Korean), <https://accesson.kisti.re.kr>

The Korea OA Journal Service in the KCI-registered Journal Portal, operated by the National Research Foundation of Korea (NRF) (in Korean), <https://www.kci.go.kr/kciportal/landing/main.kci>

The Open Access Korea (OAK) Repository Portal, operated by the National Library of Korea (in Korean), <https://www.oak.go.kr>

DataOn, operated by KISTI (in Korean), <https://dataon.kisti.re.kr>

The Korea Bio Data Station (K-BDS) (in Korean), <https://kbds.re.kr/>

The Korea Materials Data Station (K-MDS) (in Korean), <https://kmads.re.kr/>

ScienceOn, operated by KISTI (in Korean), <https://scienceon.kisti.re.kr/>

Korea DOI Center, run by KISTI (in Korean), <https://doi.or.kr/>

3.5 귀하의 국가는 오픈사이언스를 위하여 국제 혹은 연합 지역 차원의 정보통신(IT) 인프라를 운영하거나 후원하고 있는가?

Does your country **host or fund any federated regional or international information technology infrastructure for open science?** (ref.: (iii) 18.e, 1!!!) YES/ NO



Yes. The Korea Institute of Science and Technology Information (KISTI) joined the GLORIAD (Global Ring Network for Advanced Applications Development) project in 2005 and started to develop a 10 Gbps network with 15 countries in the field of big science. Since 2023, KISTI has participated in the Global 100-Gbps Ring Research Network, which connects Asia (Daejeon), North America (Chicago), and Europe (Amsterdam).

In addition, KISTI operates specific networks (i.e., LHC Optical Private Network and LHC Open Network Environment) dedicated to the transfer of LHC (Large Hadron Collider) experimental research data of CERN. Since 2013, KISTI has also participated in SCOAP3, a global partnership program to transition 11 top journals in high-energy physics to open access, which is currently in the fourth phase (2025-2027) of the program.

Overall, KISTI actively collaborates with regional and global ICT infrastructure associations. It is a key actor of the Global Network Advancement Group (GNA-G), which was formed by the integration of the Global Lambda Grid Integrated Facility (GLIF) and Global Network Advancement (GNA). KISTI plays a major role in the East Asia Resilient Backbone Network Collaboration (EARBN), an international research network in East Asia formed in August 2023. Furthermore, KISTI collaborates with the Asia Pacific Advanced Network (APAN), the Asia Pacific Oceania Network (APONET), and the Asia Europe Ring (AER).

3.6 귀하의 국가는 국가, 지역, 다국적 오픈사이언스 플랫폼 공유를 위한 연합전략과 인프라 사용을 최적화하기 위하여 북-남, 북-남-남, 남-남협력에 참여하고 있는가?

Is your country **involved in any North-South, North-South-South and South-South collaborations** to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g) **YES/ NO**



Yes. A Korean representative of the Science, Technology, Engineering & Innovation Policy Asia and the Pacific Network (STEPAN) from the Science and Technology Policy Institute (STEPI) has served as a vice chair of STEPAN from 2021 to 2024 and promoted the implementation of open science in Asia and the Pacific regions.

The Korea Institute of Science and Technology Information (KISTI) provided consulting services for the installation and operation of the NTIS platform customized for Costa Rica, enabling SINCYT (Sistema de Información Nacional Ciencia y Tecnología) to operate in the country. From 2022 to 2024, it is partnering with the Kuwait Foundation for the Advancement of Science while organizing an annual Data-AI camp for Kuwaiti high school students

In the field of ecology, the Korea Long-term Ecological Research (KLTER) enhanced regional collaboration via the International Ecology School (IES). As the host country of IES changes every year and provides a portion of the operating budget, the Republic of Korea hosted IES in 2024. The Korea Long-term Ecological Research (KLTER), a member of the International Long-term Ecological Research (ILTER) network, organized the International Ecology School (IES) in collaboration with the Environment Ecology Meteorology ICT Consilience Forum (EEMICF) in 2024; it provided relevant training to young researchers from 10 countries in the Asia-Pacific region (Korea, China, Japan, Australia, Mongolia, Taiwan, Philippines, Thailand, Vietnam, and Malaysia) in Korea.

REFERENCES:

The Science, Technology, Engineering & Innovation Policy Asia and the Pacific Network (STEPAN) (in English), <https://stepan.org>

The International Ecology School (IES), organized by the Korea Long-term Ecological Research (KLTER) and the Environment Ecology Meteorology ICT Consilience Forum (EEMICF) (in Korean), <https://www.eemicf.org/>

4. 오픈사이언스 인력, 교육훈련, 역량강화

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 귀하의 국가의 경우, 고등교육기관이나 연구수행기관에서 오픈사이언스에 관한 체계적 역량강화가 이루어지고 있는가?

Has **systematic capacity building on open science** taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c) YES/ NO



Yes. The Departments of Library and Information Science at Korean universities began to incorporate teaching about open access publishing, research data management, and open science into their curricula. The National Library of Korea also organizes regular workshops for capacity building of OAK (open access) repositories and provides educational resources on open science to librarians and information service personnel in research organizations, schools, and public institutions.

The Korea Institute of Science and Technology Information (KISTI) operates the Scientific Data Education Center, which provides data literacy and capacity-building training. It also helps users develop and utilize simulation software through the computational science and engineering simulation platform, EDISON.

Plus, the Korea Institute of Human Resources Development in Science and Technology (KIRD) offers training courses on basic research ethics, research management, and technological trends to researchers in the field of science and technology; in particular, KIRD provides educational resources on open science as part of its education on technological trends and research ethics.

REFERENCES:

The National Library of Korea, Education Website for Librarians (in Korean) : <https://edu.nl.go.kr>

The Scientific Data Education Center of the Korea Institute of Science and Technology Information (KISTI) (in Korean): <https://kacademy.kisti.re.kr>

KISTI's Computer Science and Engineering Simulation Platform, EDISON: <https://edison.re.kr>

The Korea Institute of Human Resources Development in Science and Technology (KIRD), Education

Program (in Korean): <https://alpha-campus.kr>

4.2 귀하의 국가에는 오픈사이언스의 원칙과 핵심가치를 STI 정책과 전략에 결합하는 방법에 관하여 안내하는 **정책결정자를 위한 역량강화 프로그램**이 있는가?

Have **capacity building programmes for policy makers** on how to integrate core values and principles of open science in STI policies and strategies taken place in your country? YES/ NO



Partly yes. In the process of reflecting the principles and core values of open science in major STI policies and strategies, such as the Basic Plan on Science and Technology (2023-2027), the Basic Plan on the Management and Utilization of Research Outputs (2021-2025), the Strategy for Sharing and Utilizing Research Data, and the legislative bills on Promotion of National Research Data Management and Utilization, a series of formal and informal seminars and workshops have been organized and held in order to strengthen common understanding and capabilities of policymakers. Yet, there are no long-standing educational programs on open science specifically designed for policymakers.

4.3 귀하의 국가에서 **오픈사이언스 역량에 관한 프레임워크를 고등교육 연구훈련 커리큘럼에 결합**시켰거나 혹은 2025년 말까지 결합할 예정인가?

Has **a framework of open science competencies been incorporated into higher education research skills curricula** in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b) YES/ NO



Partly yes. The curricula of library and information science departments at Korean universities began to incorporate teaching about open access publishing, research data management, and open science in general.

4.4 귀하의 국가에서 2019년 오픈교육자료 권고에서 명시한 바와 같이 2019년 정의된 **오픈교육자료**가 **오픈사이언스 역량강화의 수단으로 활용**되는가?

Are **open educational resources** as defined in the 2019 Recommendation on Open Educational Resources, **used as an instrument for open science capacity building** in your country in line with

the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

YES/ PARTLY/ NO



No. While the efforts to implement UNESCO's Recommendation on Open Educational Resources (OER) are ongoing, they are not directly linked to the capacity-building activities for open science. The Ministry of Education supports educational activities with open educational resources, whereas open science practices are primarily supported by the Ministry of Science and ICT (MSIT).

4.5. 서로 다른 학문 분야 연구자나 의사결정자, 더 나아가 일반 대중에게까지 과학지식을 전파하려는 오픈사이언스의 원칙과 가치에 따라 귀하의 국가에서 이루어진 **오픈사이언스 커뮤니케이션 지원 이니셔티브**가 있었는가?

Have there been **any initiatives** in your country **to support science communication** in line with open science values and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e) YES/ NO



Partly yes. In accordance with the Basic Plan on Science and Technology (2023-2027), the Korean government is supporting the expansion of science culture programs led by the private sector or local entities, the interactions between scientific communities and policymakers, and communications between scientists and the public. In particular, the Ministry of Science and ICT (MSIT) supports the Korea Foundation for the Advancement of Science & Creativity (KOSAC) in operating ScienceAll as a medium for science communication. In addition, there exist several other citizen science platforms, including Nature Ring in the field of biodiversity.

Furthermore, the Korean government released the 3rd Comprehensive Plan for Solving Social Problems Based on Science and Technology (2023-2027) and has made efforts to increase the contribution of national R&D programs to solving social problems. Nonetheless, no dedicated science communication project is exclusively focused on open science.

REFERENCES:

The Ministry of Science and ICT (MSIT) (February 22, 2023) The 3rd Comprehensive Plan for Solving Social Problems Based on Science and Technology (2023-2027), The Deliberative Council of the Presidential Advisory Council on Science & Technology (PACST).

ScienceAll website, operated by the Ministry of Science and ICT (MSIT) and the Korea Foundation for the Advancement of Science & Creativity (KOSAC) (in Korean), <https://www.scienceall.com>

Nature Ring (in Korean), <https://www.naturing.net/landing>

Social problem-solving R&D website in the National Science & Technology Information Service (NTIS) (in Korean), <https://www.ntis.go.kr/scisoplatform/>

5. 오픈사이언스 인센티브, 평가체계와 문화

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1 국가 혹은 기관(예: 대학, 연구수행기관 등) 차원에서 오픈사이언스 원칙과 가치에 부합하기 위하여 현재의 연구평가 과정을 검토하는 이니셔티브가 있는가? 혹은 2025년 말까지 예상되는 것이 있는가?

Have there been **any initiatives at the national or institutional level**, e.g. by universities and/or research performing institutions, **to review the existing research assessment and evaluation processes** in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c. gflj) YES/ No



Partly yes. In accordance with the Act on the Performance Evaluation and Management of National Research and Development (R&D) Programs, the government is continuously reviewing R&D results and improving evaluation methods in order to efficiently manage and utilize R&D outputs in the field of science and technology. As part of this, the 4th Basic Plan on National R&D Performance Evaluation (2021-2025) is currently being implemented. In order to overcome the shortcomings of previously dominant quantitative evaluation of publications, the government tends to emphasize qualitative evaluation of substantial research outcomes and process evaluation. However, no distinct evaluation mechanism is grounded in open science principles in existing national R&D programs.

REFERENCE:

The Act on the Performance Evaluation and Management of National Research and Development Programs

5.2 타 연구자들과나 과학계 너머의 사회 구성원들과 협력·교류하고 공유하는 등의 **오픈사이언스 실천**에 대하여 인정해주는 특정한 항목이 연구경력 평가나 승진 체계에 포함되어 있는가? 혹은 2025년 말까지 포함될 예정인가?

Have **specific clauses that recognize open science practices**, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been **incorporated in the career evaluation and progression systems** or are foreseen by the end of 2025? (ref.: (v) 20.a. 20.c. 20.d) YES/ NO



Partly yes. The 4th Basic Plan on National R&D Performance Evaluation (2021-2025) recommends that social problem-solving R&D be evaluated by the alternative indicators on social impacts beyond academic outcomes. This encourages members of society beyond the scientific community to participate in the R&D evaluation process. Furthermore, public research institutions and government-affiliated agencies are required to provide their data and materials produced by public funds to societal members in a proactive manner under the Open Government Data policy; a public organization's performance on open government data is annually reviewed and evaluated. Nevertheless, it is not common to recognize and reward scientists' efforts to collaborate with societal members outside a scientific community in the existing career development and promotion systems in science.

REFERENCE:

The Ministry of Science and ICT (MSIT) (August 26, 2020) The 4th Basic Plan on National R&D Performance Evaluation (2021-2025), The Deliberative Council of the Presidential Advisory Council on Science & Technology (PACST).

5.3 귀하의 국가에서 **오픈사이언스 실천을 인정해주고 보상해주는 이니셔티브**가 지금까지 있었는가? 혹은 2025년 말까지 예상되는 이니셔티브가 있는가?

Have there been **any initiatives** in your country **to recognize and reward open science practices** or are they foreseen by the end of 2025? (ref.: (v) 20.a. 20.b, 20.c, 20.e. 20.f. 20.g, 20h)



Yes. The Minister of Science and ICT annually confers the Knowledge Sharing Award upon five outstanding researchers who have created and shared excellent academic achievements based on

information collected from the National Science & Technology Information Service (NTIS). In addition, the Minister of Culture, Sports and Tourism, through the National Library of Korea, has been presenting the Open Access Korea Merit Award since 2015 to individuals and organizations that have contributed to the expansion of open access movement.

5.4 귀하의 국가에서 오픈사이언스 실천의 예기치 않은 부정적 결과가 보고된 적이 있는가?

Have there been **any unintended negative consequences of open science practices** reported in your country? (ref.: (v) 20) YES/ NO



Yes. Along with the growth of open access journals, the problems of predatory journals and conferences intensified. In order to prevent the negative consequences, the Center for Research Ethics (CRE) was established at the National Research Foundation of Korea (NRF) in 2020. The Korea Institute of Science and Technology Information (KISTI) also developed a system to identify predatory journals and operates a website, Scholarly Ecosystem Against Fake Publishing Environment (SAFE).

REFERENCES:

The Center for Research Ethics (CRE) of the National Research Foundation of Korea (NRF) (in Korean), <https://cre.nrf.re.kr/>

Scholarly Ecosystem Against Fake Publishing Environment (SAFE) website, operated by the Korea Institute of Science and Technology (KISTI) (in Korean), <https://safe.accesson.kr/>

6. 오픈사이언스 모범사례 및 혁신사례

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT

STAGES OF THE SCIENTIFIC PROCESS

6.1 국가 혹은 기관 차원에서 연구 과정 전체의 개방성을 높이기 위하여 연구 사이클의 서로 다른 단계에서 혁신적이거나 참여적인 방법과 절차의 변화를 도모하는 이니셔티브들이 있었는가?

Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21) YES, at both national and institutional levels.



Yes. There is a case of open research agenda-setting that occurred at the beginning of the research. In addition, the Ministry of Science and ICT (MSIT) announced a plan to improve national R&D projects and related policies through the open participation of domestic academic societies at the "2023 General Meeting of Academic Societies for R&D Innovation."

Since 2005, the Korean government has required research facilities over a certain size purchased through national R&D projects to be registered to the National Research Facilities & Equipment Center (NFEC) and the registered information to be available to the public so that other researchers outside the project team can identify and utilize the registered facilities.

ACOMS+ is a platform for article submission and review developed by the Korea Institute of Science and Technology Information (KISTI); it also supports open peer review for domestic journals. If a submitted paper is decided to go through open peer review, its review information becomes available upon publication.

REFERENCES:

The National Research Facilities & Equipment Center (NFEC)'s website on research facility and equipment information service, ZEUS (in Korean), <https://www.zeus.go.kr/main>

The Korea Institute of Science and Technology Information (KISTI)'s ACOMS+ (in Korean), <https://acoms.accesson.kr>

7. 오픈사이언스 국제협력·다자협력

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF

OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 2021년 오픈사이언스 권고에서 제시한 바와 같이 귀하의 국가는 **오픈사이언스에 관한 국제과학협력**을 어떻게 촉진하고 있는가?

How is your country promoting and facilitating **international scientific collaborations on open science** as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)



Yes. The Republic of Korea is fully engaged in the process of shaping and implementing the UNESCO Recommendation on Open Science, including its involvement in the UNESCO Open Science Advisory Council from the early stage. The Korean government is expanding international cooperation on open science through multilateral meetings such as the OECD, G20, and APEC, joint committee meetings with partner countries, and high-level summits.

At the institutional level, the National Research Foundation of Korea (NRF) collaborates with the Global Research Council (GRC) while discussing open science agenda. Moreover, in response to COVID-19, the Korea Disease Control and Prevention Agency (KDCA) participated in GISAID's data sharing. Korean researchers and experts are actively developing field-specific research collaboration networks across countries, and GloPID-R's APIS (Asia-Pacific Infectious Disease Shield) is one such example. Korean actors are also developing relevant regional Asia-Pacific networks, which include COAR (Confederation of Open Access Repositories) Asia, APRI (Asia Pacific Research Integrity), APAN (Asia Pacific Advanced Network), and STEPAN (Science, Technology, Engineering & Innovation Policy Asia and the Pacific Network). As part of APAN activities, the Korea Institute of Science and Technology Information (KISTI) is expanding partnerships with NII (National Institute of Informatics) of Japan and CSTCloud of China. KISTI is also enhancing its partnership with the Japan Science and Technology Agency (JST) and the Institute of Scientific and Technical Information of China (ISTIC), hosting a joint seminar on science and technology information every year. The National Library of Korea collaborates with the International Standard Name Identifier (ISNI) and the Virtual International Authority File (VIAF); it hosted an international event, the Dublin Core Metadata Initiative (DCMI), at Kyungpook National University in 2022.

Regarding research data, Korean actors collaborate with RDA, CODATA, and WDS, whereas they work with COAR, OA2020, COALition S, and ICOLC for open access. In 2022, the Ministry of Science and ICT (MSIT), KISTI, the National Library of Korea, the National Assembly Library, the Korea Research Institute of Standards and Science, Sungkyunkwan University, the Seoul Metropolitan Government, the Korea Institute of Oriental Medicine, and the Korea Institute of Geoscience and

Mineral Resources jointly hosted the International Data Week (IDW), a biennial event co-organized by RDA, CODATA, and WDS.

The Republic of Korea has worked closely with UNESCO networks and entities for open science. From the beginning of the recommendation formation process, a Korean representative was involved in the UNESCO Open Science Advisory Council as a regional representative; even after the endorsement of the Recommendation, Korean experts have continued to participate in the five working groups on open science led by the UNESCO headquarters (secretariat). Korean actors also interact with UNESCO's regional offices (Jakarta, Beijing, etc.). As the vice chair of STEPAN, a Korean representative from the Science and Technology Policy Institute (STEPI) promotes open science implementation in the region. On top of that, KISTI recently initiated an international joint research project (scheduled from 2024 to 2029) with UNESCO headquarters to study how to vitalize global open science.

7.2 귀하의 국가는 오픈사이언스를 위한 모범사례의 공유, 역량강화, 평가지표 등에 관한 사항을 포함하여 오픈사이언스 관련 초국경 다중이해관계자 협력을 촉진하기 위한 전략이나 계획을 가지고 있는가?

Does your country have **a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science**, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f) YES/ NO



Partly yes. The Korean government is implementing a national strategy for international STI cooperation, not limited to the promotion of open science. Specifically, in 2023, the Yoon administration proposed the "Strategy for Global R&D Leading the World" and the "R&D Innovation Strategy: Leaping to a Global Science and Technology (S&T) Hub Leading the World." Based on those strategies, the government is promoting global research collaboration, particularly in the field of critical and emerging technologies, and enhancing research environments aligned with global standards. As a result, cross-border multi-stakeholder collaborations related to open science are indirectly being fostered, even if they are not explicitly stated as open science collaborations.

REFERENCES:

Joint Interagency Government (November 27, 2023) A Strategy for Global R&D Leading the World, The Plenary Council of the Presidential Advisory Council on Science & Technology (PACST).

Joint Interagency Government (November 27, 2023) Yoon Seok-yeol Administration's R&D Innovation Strategy: Leaping to a Global Science and Technology (S&T) Hub Leading the World, The Plenary Council of the Presidential Advisory Council on Science & Technology (PACST).

7.3 귀하의 국가는 오픈사이언스를 위한 국제적·지역적 펀딩 메커니즘의 설치에 관한 논의에 참여하고 있는가?

Is your country **involved in any discussion on the establishment of regional and international funding mechanisms for open science?** YES/ NO



Yes. Korean experts continue participating in international meetings on open science funding as part of the UNESCO Open Science Working Groups.

0. 응답자 기본정보

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Full Name:
- Position:
- Organization(s) or entity(ies) (including non-governmental) consulted for completing this survey:

Name of the organization*:

Website*:

Email address*

Sector: ☐ Public ☐ Private ☐ Civil Society ☐ Other